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Optogenetic stimulation of the locus coeruleus enhances appetitive extinction in rats

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Abstract

Extinction is a specific example of learning where a previously reinforced stimulus or response is no longer reinforced, and the previously learned behaviour is no longer necessary and must be modified. Current theories suggest extinction is not the erasure of the original learning but involves new learning that acts to suppress the original behaviour. Evidence for this can be found when the original behaviour recovers following the passage of time (spontaneous recovery), or reintroduction of the reinforcement (i.e., reinstatement). Recent studies have shown that pharmacological manipulation of noradrenaline (NA) or its receptors can influence appetitive extinction, however, the role and source of endogenous NA in these effects is unknown. Here, we examined the role of the locus coeruleus (LC) in appetitive extinction. Specifically, we tested whether optogenetic stimulation of LC neurons during extinction of a food-seeking behaviour would enhance extinction evidenced by reduced spontaneous recovery in future tests. LC stimulation during extinction trials did not change the rate of extinction but did serve to reduce subsequent spontaneous recovery suggesting that stimulation of the LC can augment reward-related extinction. Optogenetic inhibition of the LC during extinction trials reduced responding during the trials where it was applied, but no long-lasting changes in the retention of extinction were observed. Since not all LC cells expressed halorhodopsin, is possible that more complete LC inhibition or pathway-specific targeting would be more effective at suppressing extinction learning. These results provide further insight into the neural basis of appetitive extinction, and in particular the role of the LC. A deeper understanding of the physiological bases of extinction can aid development of more effective extinction-based therapies.

eLife assessment

In this **important** study, Lui and colleagues examine whether the locus coeruleus is involved in extinction of an appetitive conditioned response. Using a set of optogenetic approaches aimed at manipulating the activity of locus coeruleus cells, the authors provide **solid** evidence that these neurons regulate the extinction of conditioned responses. Overall this study further highlights the key role of noradrenaline in cognitive processes and will be of interest to those interested in associative learning, extinction, noradrenaline, associated brain systems and translational endpoints.

Introduction

The ability to detect and encode relationships between events and modify previously learned behaviours when confronted with new information is essential for adaptive, flexible control of behaviour. Extinction is a specific example where a previously reinforced stimulus or response is no longer reinforced, and the previously learned behaviour is no longer necessary. Contemporary theories of learning suggest that extinction is not simply the erasure of previous learning but rather involves new inhibitory learning, suppressing the original learned behaviour (Quirk and Mueller 2007; Bouton et al., 2021; Iordanova et al., 2021; McNally et al., 2021). Evidence for this comes from numerous demonstrations that the original learning can be uncovered following reintroduction of the original reinforcer (i.e., reinstatement; Rescorla & Heth, 1975), testing outside the extinction context (i.e., renewal; Bouton & King, 1980; Garcia-Gutierrez & Rosas, 2003; Nelson et al., 2011), or simply by allowing the passage of time between extinction training and testing (i.e., spontaneous recovery; Rescorla, 1996, 2004). These phenomena mean that extinction-based therapies are not always successful in suppressing previously-established behaviours, thus sparking interest in developing methods for augmenting extinction learning. More complete understanding of the neural and pharmacological bases of extinction may improve progress in this regard.

Recent studies have shown that noradrenaline (NA) can influence extinction learning. For example, blocking NA reuptake with atomoxetine can augment extinction of reward-related behaviours (Janak and Corbit, 2011; Janak et al., 2012; Furlong et al., 2015; Leung & Corbit, 2017). In contrast, the application of the beta-receptor antagonist propranolol impairs extinction learning related to rewarding (Janak and Corbit, 2011; Janak et al., 2012; Leung & Corbit, 2017) and fear-provoking events (Berlau & McGaugh, 2006; Do-Monte et al. 2010; Mueller et al., 2008). Consistent with a role in extinction, experiments using fast-scan voltammetry observed a surge in NA release that coincided with the omission of an expected reward implicating the endogenous noradrenergic system in detecting extinction contingencies (Park et al., 2013).

While these studies illustrate that NA can enhance the strength or longevity of extinction learning, the source of NA mediating these effects is unknown. It is likely that projections originating in the locus coeruleus (LC) are responsible as the LC has direct anatomical connections with multiple brain regions that contribute to extinction including the basolateral amygdala (BLA) and the infralimbic cortex (IFC) (McCall et al., 2017; Uematsu et al., 2017). Previous studies that recorded LC activity in freely behaving rats observed that LC activity increased when rats were presented with novel stimuli, and while this response rapidly habituated, LC activity rapidly increased again when a stimulus was followed by reward (Sara et al., 1994), suggesting the LC responds to novelty. However, the LC was also reported to increase activity when stimulus-reinforcer contingencies were reversed, and during extinction (Sara & Segal, 1991; Sara et al., 1994; Bouret and Sara, 2004), suggesting that the LC may not simply be responding to novelty, but also to changes in expected outcomes in order to facilitate a corresponding change in behaviour.

Looking back to earlier studies that directly tied LC manipulations to extinction learning, lesions of the LC were shown to impair extinction of conditioned fear (Mason et al., 1978) and 6-OHDA lesions targeting the dorsal noradrenergic bundle slowed extinction in an appetitive runway task (Mason & Iverson, 1975) although electrolytic lesions of the LC, which deplete forebrain NA less completely, were found to have little impact on appetitive extinction (Koob et al., 1978). While more recent studies using either optogenetic or chemogenetic methods to activate the LC have reported somewhat conflicting results, in some cases enhancing and other cases disrupting extinction (Uematsu et al., 2017; Giustino et al., 2019; 2020), this could be explained by differences in levels of fear, differences in the intensity of stimulation, as strong LC stimulation can be fear promoting, or by an increasing appreciation of the modular organization of the LC and differential role of different projection targets (Uematsu et al., 2017; Chandler et al., 2019). It is important to note that recent studies have largely focused on the extinction of conditioned fear and relatively less is known about whether LC stimulation can modulate extinction of reward-related behaviours. This distinction is not trivial because fear conditioning paradigms also inherently involve noxious stimulation, the impact of which can be amplified by LC stimulation, and these paradigms involve a degree of stress or anxiety, conditions tied to NA levels (e.g., Chiang & Aston-Jones, 1993; Giustino et al., 2019; Uematsu et al., 2017).

While electrophysiological recording data show that LC neurons respond to sudden reward omission (Sara & Segal, 1991; Sara et al., 1994; Su & Cohen, 2022), these results have been inconsistent (Bouret & Sara, 2004). It remains unclear whether any change in activity reflects a response to novelty or change, or contributes to learning a new extinction contingency. The goal of the following study was to examine the role of the LC in appetitive extinction. Specifically, we examined whether direct optogenetic stimulation of LC neurons during extinction of a food-seeking behaviour would enhance extinction evidenced by reduced spontaneous recovery in future tests.

Results

Histological Verification of LC-NA neurons across treatment groups

Rats were trained in a discriminated operant task, where lever-pressing during stimulus presentations resulted in food reward. Responding was then extinguished across three sessions and retention of extinction tested one and seven days later. Schematics of training, testing and optogenetic manipulation are provided in **Figures 2A** and **4D**. In the first experiment, 43 male and female rats were divided into three treatment groups: A channelrhodopsin-2 (ChR2) group, which consisted of tyrosine hydroxylase (TH)-Cre rats injected with AAV5-EF1a-DIO-ChR2(H134R)-mCherry, an Offset group, which consisted of TH-Cre rats injected with the same virus expressing ChR2 but received stimulation during the inter-trial intervals (ITI), and Controls, which consisted of TH-Cre Long-Evans (LE) rats (n=7, receiving light pulses during the CS) and wild-type LE rats (receiving light pulses during CS presentations; n=8, or during the ITI; n=7) injected with the control virus (AAV5-EF1a-DIO-mCherry). Preliminary analyses demonstrated no differences between TH-Cre and wild-type controls (stimulus and offset) during training [**Fig. 1E**; $F(2,19)=1.149$, $p=0.338$], extinction [**Fig. 1F**; $F(2,19)=1.883$, $p=0.179$], or testing [**Fig. 1G**; Test 1: $F(2,19)=0.352$, $p=0.708$]; Test 2: $F(2,19)=0.250$, $p=0.781$] and so they were collapsed for further analyses and are henceforth referred to as the control group.

Cell counts and calculation of infection rates were conducted on sections of the LC positioned between -9.68 and -10.04 mm in the anterior-posterior plane (Paxinos and Watson, 2007, 6th Edition). Each section provided ~116 ± 25 TH-positive LC neurons as determined by DAPI and TH/Alexafor-488 staining. Viral expression was identified by colocalized TH/AlexFluor 488 and mCherry expression. Animals that did not have viral expression in the LC were not included in the experimental groups. TH-Cre LE rats showed clear viral expression in the LC with approximately 73±2.8% of all TH+ cells in the LC expressing mCherry, and 97 ± 2.5% of infected cells staining positive for TH indicating robust and selective targeting of noradrenergic LC neurons (**Fig. 1A**). Males and females showed very similar infection rates (Males, 74%; Females, 72%). Unsurprisingly, wildtype LE rats lacked Cre expression and therefore had no mCherry expression (not shown). Only animals with optic implants bilaterally within the LC were included in behavioural analyses (**Figure 1B-D**).

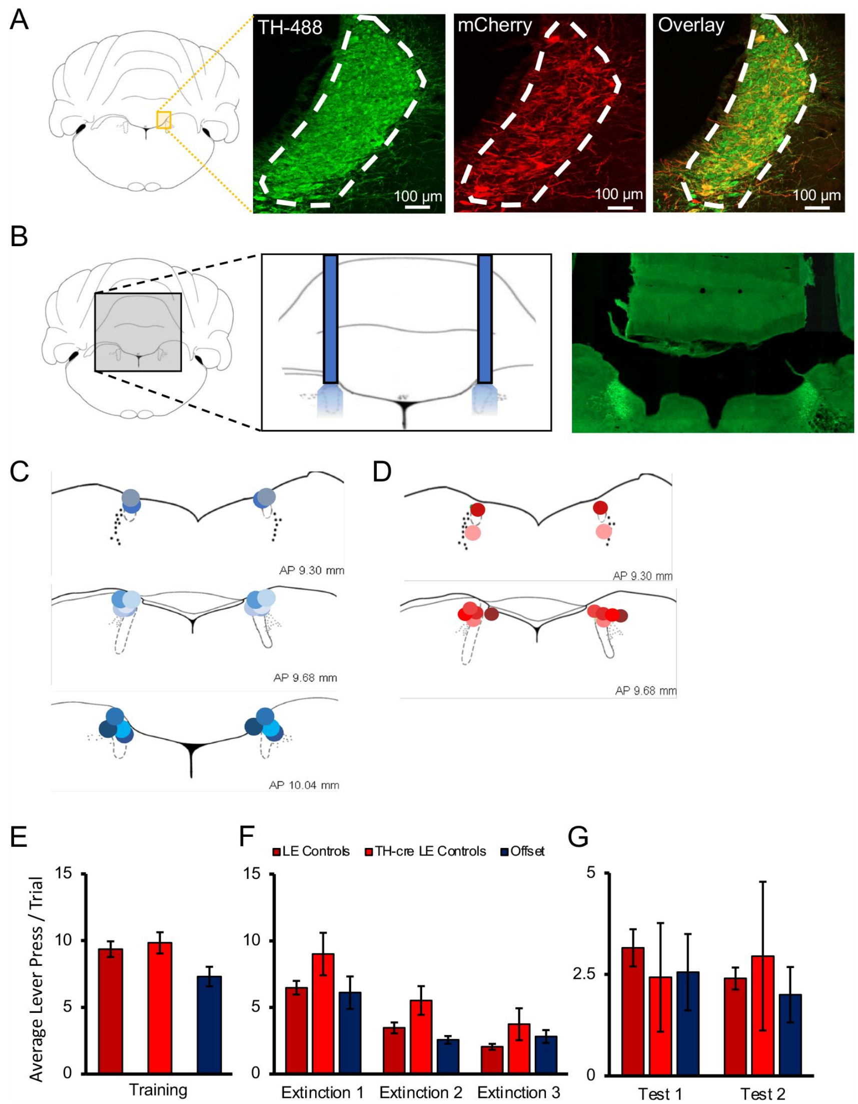

Figure 1.

Histological verification of optogenetic targeting of noradrenergic LC neurons.

(**A**) To verify viral expression, the locus coeruleus (LC) was stained with tyrosine hydroxylase (TH) primary antibody along with secondary antibody with AlexaFluor 488 to identify TH-positive neurons in the region of the LC. The same tissue was counterstained with mCherry primary antibody and enhanced with AlexaFluor 568 to determine the overlap of virally infected TH-positive cells. Cell counts and calculation of infection rates were conducted on sections of the LC positioned between -9.68 and -10.04 mm in the anterior-posterior plane (Paxinos and Watson, 2007, 6th Edition). Each section provided ~116 ± 25 TH-positive LC neurons as determined by DAPI and TH/AlexFluor 488 staining. Viral expression was identified by colocalized TH and mCherry expression. Animals that did not have viral expression in the LC were not included in the experimental groups. TH-Cre LE rats showed clear viral expression in the LC with approximately 73±2.8% of all TH+ cells in the LC expressing AAV5-EF1a-DIO-ChR2(H134R)-mCherry, with an example provided in the overlay, and 97 ± 2.5% of infected cells staining positive for TH indicating robust and selective targeting of noradrenergic LC neurons (**Fig. 1A**). Males and females showed very similar infection rates (Males, 74%; Females, 72%). (**B**) The approximate probe tip locations in each animal are displayed confirming successful targeting of the LC, with shades of blue representing animals expressing ChR2 (**C**), and shades of red representing control animals infected with AAV5-EF1a-DIO-mCherry (**D**). When the behaviour of multiple control groups was compared, TH-Cre and LE rats expressing control virus and receiving light delivery during stimulus presentation in extinction, as well as Offset controls expressing ChR2 virus and receiving light delivery during the ITI of extinction sessions showed no significant differences in response rates on the final day of training (**E**), during extinction sessions with light delivery (**F**), or during tests of spontaneous recovery conducted without stimulation the next day (Test 1) and again one week later (Test 2; (**G**; p>0.05). These multiple control groups are therefore collapsed in subsequent analyses.

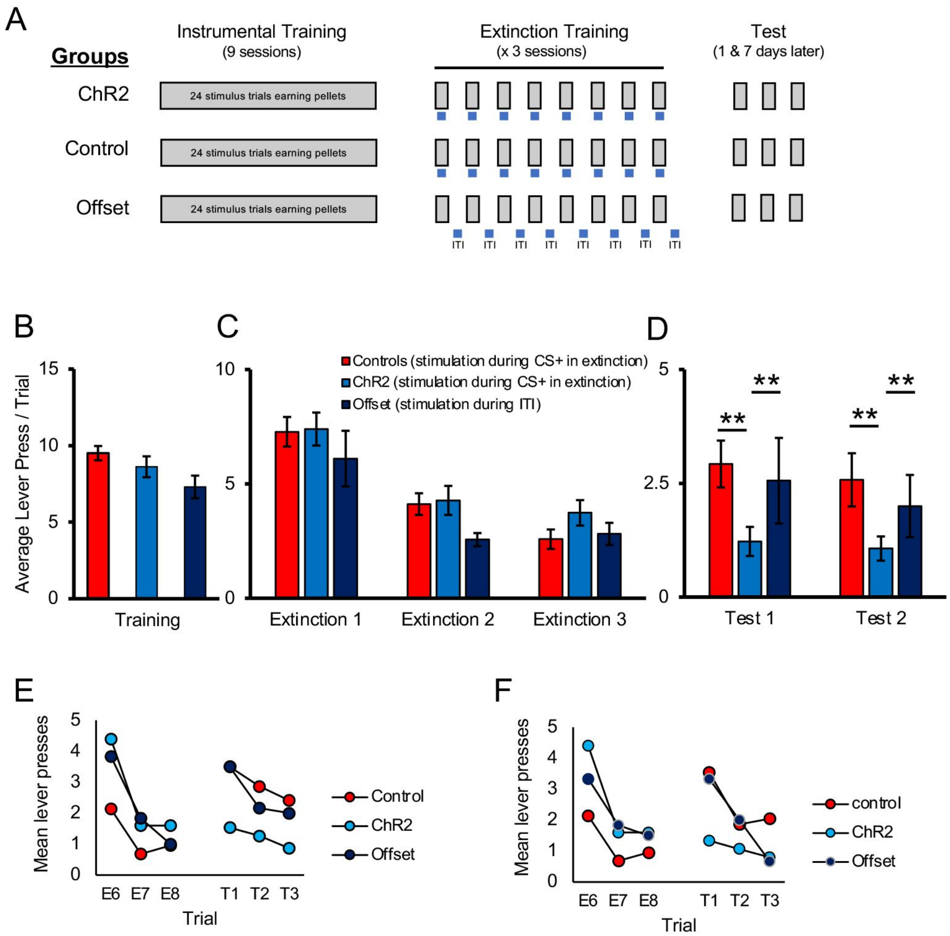

Figure 2.

Optogenetic stimulation of the LC enhances long-term expression of extinction.

(A) Schematic representation of the study design. All groups received identical discriminated operant training where lever-pressing during 20-s presentations of a discriminative stimulus was reinforced with a food pellet. Responding in the absence of the discriminative stimuli was not reinforced. The groups differed in the period of time in which optical manipulation of the LC took place during extinction sessions with the timing of light delivery indicated by the blue bars; for Control and ChR2 groups light was delivered during stimulus presentations (gray bars) when reward was expected based on previous training, but not delivered; the Offset group received similar LC stimulation but offset from stimulus presentations (i.e., during the ITI). The 3 groups (Control, ChR2, and Offset groups), showed similar response rates in training (**B**; p>0.05). Responding in all groups extinguished when reward was omitted (p<0.05) and no differences were observed between groups (**C;** p>0.05). When tested the next day for spontaneous recovery, the ChR2 group displayed significantly fewer lever presses than the Control or Offset groups, suggesting LC stimulation strengthened extinction learning (**D**, Test 1; p<0.05). A similar effect was observed one week later indicating a persistent effect of LC stimulation on later retention of extinction (**D**, Test 2; p<0.05). **(E, F)** To directly test for spontaneous recovery, we compared responding during the last trial of extinction to the first test trial for each group. In Test 1 the control group increased responding from the end of extinction to the beginning of the test the following day (**E**, p<0.001), thus demonstrating spontaneous recovery. A similar pattern was observed in the offset group, although this effect was marginal (p=0.055). Of note, there was no change in responding across time in the ChR2 group (p=0.890) and responding remained low in this group indicating suppressed spontaneous recovery. To test whether this was a lasting effect, we tested the same rats again one week later where we observed a similar effect of group [Test 2; **Fig. 2D**; F(2,40)=3.732, p=0.033] resulting again from reduced responding in the ChR2 group compared to the Control [t(35)=2.663, p=0.006] or Offset groups [t(19)=2.117, p=0.024], which did not differ [t(26)=0.506, p=0.309]. To demonstrate spontaneous recovery, trial data were again considered. The control group increased responding from the end of extinction to the beginning of the test conducted one week later (**F**; p=0.002) thus demonstrating spontaneous recovery. A similar pattern was observed in the offset group (p=0.049). Of note, there was no change in responding across time in the ChR2 group (p=0.256) suggesting that LC stimulation can enhance extinction learning, increasing its resilience against spontaneous recovery.

Photostimulation of ChR2-expressing LC cells enhances extinction learning

A schematic of training and testing procedures is provided in **Figure 2A**. We first examined the effect of stimulating noradrenergic LC neurons during extinction. Rats were trained in a discriminated operant task where lever-pressing during 20-s stimulus presentations resulted in delivery of a food reward whereas responses in the absence of a stimulus had no programmed consequences. At the end of acquisition, TH-Cre rats expressing ChR2 were assigned to either the experimental group (ChR2), that received optogenetic stimulation (5-ms pulses at 10 Hz) for the duration of each stimulus presentation across trials in the upcoming extinction sessions or the Offset group that received equivalent stimulation during the ITI. These groups were compared to the Control group that was administered a control virus (AAV5-EF1a-DIO-mCherry) and so equivalent optogenetic stimulation was expected to be without effect. On the final day of acquisition, there were no group differences in response rates (lever presses per trial) during stimulus presentations [**Fig. 2B**; $F(2,40)=2.302$, $p=0.113$]. As illustrated in **Figure 2C**, lever-pressing rates decreased across days of extinction [$F(2,80)=52.686$, $p=0.001$] and there was no main effect of group [$F(2,40)=0.934$, $p= 0.402$] or interaction between day and group [$F(4,80)=1.061$, $p=0.381$] suggesting that optogenetic LC stimulation did not have an immediate effect on instrumental performance during extinction training. However, we hypothesized that LC stimulation may still influence retention of extinction learning. To examine this, we tested for spontaneous recovery of responding the next day, without stimulation. In this test, there was an effect of group [Test 1; **Fig. 2D**; $F(2, 40)=4.037$, $p=0.025$] and pairwise comparisons demonstrated that rats in the ChR2 group responded significantly less compared to control [$t(35)=2.9.04$, $p=0.003$] or offset [$t(19)=1.735$, $p=0.049$] groups. The Control and Offset groups did not differ from each other [$t(26)=0.522$, $p=0.303$]. While the test data clearly show lower responding in the ChR2 group, responding overall was low at test and comparison of the final extinction session to the test sessions does not show clear evidence of spontaneous recovery in controls. One reason for this is that the extinction sessions were longer than the tests and relatively high responding at the beginning of the session was averaged with relatively lower responding in latter trials. To illustrate spontaneous recovery, we compared the final three trials of extinction to the first three trials of testing (**Figure 2E**, 2F). Analysis of the trial data confirmed an effect of trial ([$F(5, 200)= 5.655$, $p=0.001$] and while there was no main effect of group [$F(2, 40)=0.332$, $p=0.719$], there was a trial by group interaction [$F(10, 200)=3.127$, $p=0.001$] indicating that responding across trials differed by group. To directly test for spontaneous recovery, we compared responding during the last trial of extinction to the first test trial for each group. As shown in **Figure 2E**, the control group increased responding from the end of extinction to the beginning of the test the following day [$t(21)=3.745$, $p=0.001$] thus demonstrating spontaneous recovery. A similar pattern was observed in the offset group, although likely due to the smaller group size, this effect was marginal [$t(5)=1.946$, $p=0.055$]. Of note, there was no change in responding across time in the ChR2 group [$t(14)=0.445$, $p=0.890$]. As responding remained low in this group (**Figure 2E**) this suggests that LC stimulation enhanced retention of extinction making it less susceptible to disruption by the passage of time. To test whether this was a lasting effect, we tested the same rats again one week later where we observed a similar effect of group [Test 2; **Fig. 2D**; $F(2,40)=3.732$, $p=0.033$] resulting again from reduced responding in the ChR2 group compared to the Control [$t(35)=2.663$, $p=0.006$] or Offset groups [$t(19)=2.117$, $p=0.024$], which did not differ [$t(26)=0.506$, $p=0.309$]. To demonstrate spontaneous recovery, a similar analysis was performed on trial data. Comparison of the final three trials of extinction to the three test trials one week later demonstrated a significant effect of trial [$F(5, 200)=6.923$, $p=0.001$], and while there was no main effect of group [$F(2,40)=0.109$, $p=0.897$], there was an interaction between these factors [$F(10, 200)=13.154$, $p=0.001$]. As shown in **Figure 2F**, the control group increased responding from the end of extinction to the beginning of the test conducted one week later [$t(21)=3.166$, $p=0.002$] thus demonstrating spontaneous recovery. A similar pattern was observed in the offset group [t(5)=2.038, p=0.049]. Of note, there was no change in responding across time in the ChR2 group [t(14)=0.673, p=0.256]. This suggests that LC stimulation can enhance extinction learning, increasing its resilience against spontaneous recovery.

LC photostimulation does not produce a conditioned place aversion

Studies have shown that the LC is highly responsive to aversive stimulation, such as foot shocks (Chiang and Aston-Jones, 1993; Sara et al., 1994; Chen and Sara, 2007), and the NA released contributes to anxiety (Curtis et al., 2012; McCall et al., 2015). Importantly, longer duration tonic LC stimulation has been reported to induce anxiety-like behaviours and conditioned aversion (McCall et al., 2017). It is therefore possible that the reduction in response rates observed in the ChR2 group could be caused by the formation of an aversive association with the discriminative stimuli or some other aspect of the task.

It is also possible that LC stimulation could have triggered an incompatible fear state that interfered with appetitive responding rather than directly promoting appetitive extinction learning (Rescorla and Solomon, 1967). To address this, we conducted a conditioned place preference/place aversion test where rats expressing ChR2 (n=10) received LC stimulation with similar parameters to those used during extinction in one of two chambers (Context A or B, counterbalanced) alternating daily with equal exposure to a second context where no stimulation was delivered (see **Fig. 3A**). After 3 days of exposure to each context with or without LC stimulation, rats were given free access to both chambers and the time spent in each context was recorded. We found that rats spent equivalent time in the context paired with LC stimulation and in the context where no stimulation was delivered [**Fig. 3B**; t(9)= 1.199, p=0.261], suggesting the rats had developed no preference for or aversion to either context and, therefore, that the reduction in lever-pressing in tests of spontaneous recovery was unlikely due to anxiety or aversive associations produced by LC stimulation.

LC photoinhibition does not impact extinction learning

Having found that LC stimulation enhanced retention of extinction, we next examined whether LC activity was necessary for appetitive extinction by inhibiting noradrenergic LC cells during extinction training. Seventeen rats underwent surgery to introduce halorhodopsin (eNpHR; AAV5-EF1a-DIO-eNpHR3.0-mCherry; final n=7) or a control virus (AAV5-EF1a-DIO-mCherry; final n=8) into the LC, followed by the insertion of a dual-optic implant bilaterally targeting the LC. Histological assessment confirmed that 55 ± 8.7% of TH+ cells expressed mCherry, and only animals with implant tips within the LC were included in behavioural analyses; **Fig. 4A-C**. Rats were trained in an identical manner to the ChR2 experiment described above, except that LC illumination was with continuous 625 nm wavelength light to inhibit eNpHR-expressing noradrenergic LC cells during each 20-s stimulus presentation across extinction training. Following LC inhibition during extinction, the rats were tested for spontaneous recovery the next day and again one week later without LC inhibition, identical to the tests described above (**Fig. 4D**). Both groups had similar response rates at the end of acquisition [**Fig. 4E**; F(1,13)=0.126, p=0.728]. Mean response rates per stimulus decreased across days of extinction training [**Fig. 4F**; F(2,26)=43.541, p=0.001], with the eNpHR group responding less overall [F(1,13)=6.899, p=0.021] however, there was no interaction between these factors [F(2,26)=2.356, p=0.115]. We observed no differences between groups in mean response rates in extinction tests without optogenetic manipulation conducted the next day [Test 1; **Fig. 4G**; F(1,13)=0.004, p=0.949] and one week later [Test 2; F(1,13)=0.465, p=0.507].

To directly test for spontaneous recovery, we compared responding during the last trial of extinction to the first test trial for each test. As shown in **Figure 4H**, there was a significant effect of trial with responding increasing from the end of extinction to the beginning of the test the

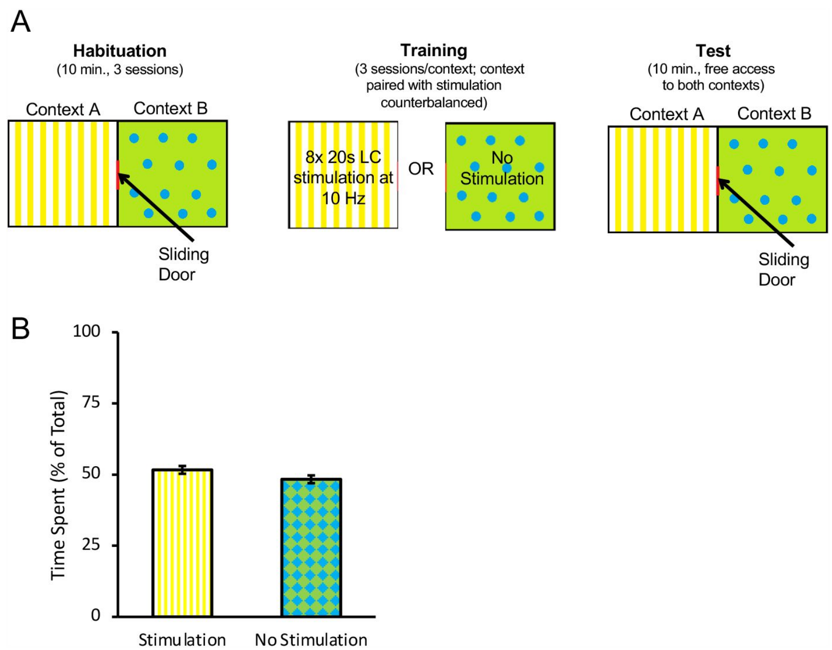

Figure 3.

Optogenetic stimulation of the LC does not produce conditioned aversion.

A conditioned place preference/aversion test was performed to determine if LC stimulation produced aversive conditioning. Rats expressing ChR2 were placed in an apparatus with two chambers separated by a sliding door, depicted in (**A**). One chamber, labelled Context A, had a yellow background with vertical stripes. A second chamber, labelled Context B had a green background with blue circles. All rats were given 3 days to freely explore both chambers for 10 min per day. Afterwards, the door was closed, and rats were confined to one context (A or B, counterbalanced) where they received eight 20 s bouts of LC stimulation at 10 Hz (160 s total stimulation time) within the 10-min session. On alternating days rats were confined to the other context (B or A, counterbalanced) without stimulation. On the test day, the sliding door opened and rats were given 10 min to freely explore. The total time spent in each context was recorded. A rat was considered to be in one chamber when both hindfeet were inside the chamber. No preference for either context was observed as rats spent a near-equal amount of time in both chambers (**B**), suggesting no aversive associations with the context formed as a result of optogenetic stimulation of the LC with the parameters used here.

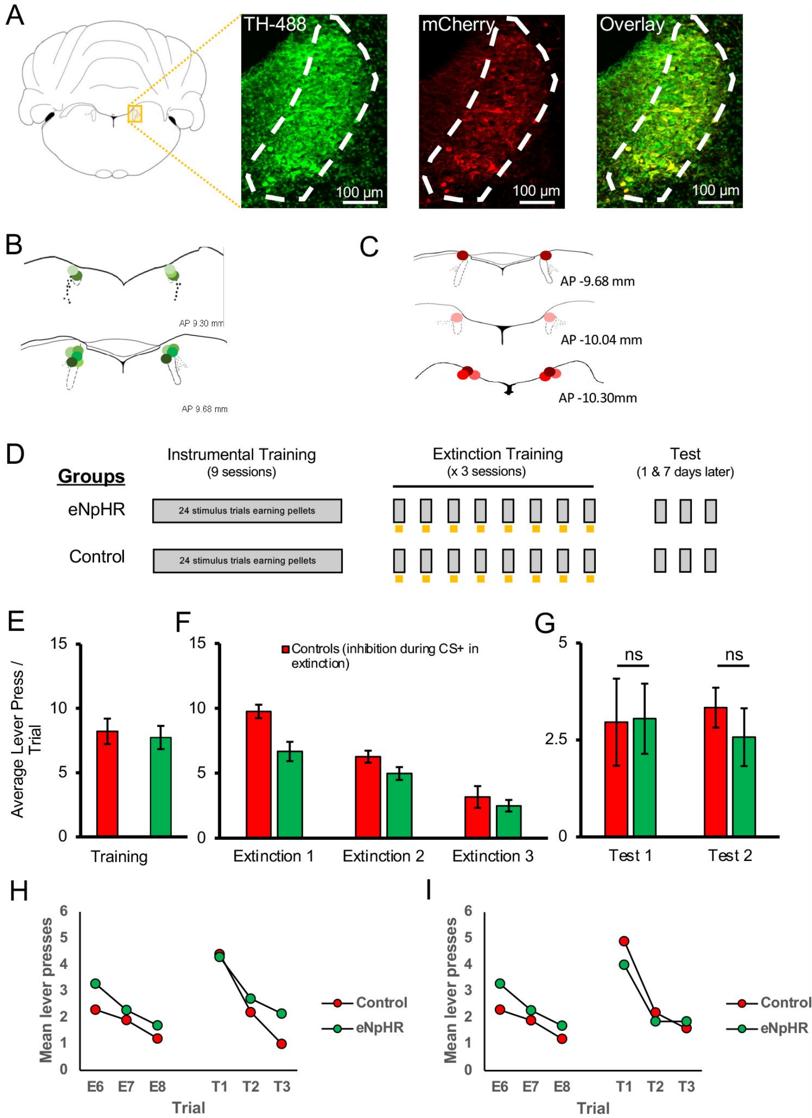

Figure 4.

Optogenetic inhibition of the LC did not impair the long-term expression of extinction.

LC neurons were successfully targeted with halorhodopsin-expressing virus (AAV5-EF1a-DIO-eNpHR3.0-mCherry) or a control virus (AAV5-EF1a-DIO-mCherry) as shown in (**A**). Green represents TH-positive neurons, red represents virally infected neurons expressing mCherry, and infected TH-positive cells are shown in the overlay (55±8.7% infection rate). The locations of probe tips are shown with shades of green representing animals expressing eNpHR (**B**), and shades of red representing control animals (**C**). (**D**) Schematic representation of training and testing. Gray bars represent stimulus presentations and yellow bars represent light delivery. Both groups underwent identical discriminated operant training and showed comparable response rates at the end of acquisition (**E**; $p>0.05$). Control and eNpHR groups were then given LC light delivery during stimulus trials over three extinction sessions. Photoinhibition of the LC during extinction training reduced mean response rates compared to control rats (**F**; $F(1,13)=6.899$, $p=0.021$) but this difference did not persist to tests conducted one (Test 1) and seven (Test 2) days later where no group differences were observed in spontaneous recovery (**G**; $p>0.05$). **(H, I)** To directly test for spontaneous recovery, we compared responding during the last trial of extinction to the first test trial for each test. There was a significant effect of trial with responding increasing from the end of extinction to the beginning of the test the following day (**H**; $p=0.020$). There was no effect of group and no trial x group interaction ($p=0.780$) suggesting spontaneous recovery was similar in the two groups. Similar results were found for the test conducted one week later where responding increased from extinction to testing thus demonstrating spontaneous recovery (**I**; $p=0.011$). But again there was no effect of group or trial x group interaction. Together these results suggest that spontaneous recovery was observed retention of extinction was equivalent between groups.

following day [F(1,15)=6.828, p=0.020]. There was no effect of group [F(1,15)=0.030, p=0.865] and no trial x group interaction [F(1,15)=0.081, p=0.780] suggesting spontaneous recovery was similar in the two groups. Similar results were found for the test conducted one week later where we observed a significant effect of trial, with responding increasing from extinction to testing thus demonstrating spontaneous recovery [F(1,15)=8.527, p=0.011]. But again there was no effect of group [F(1,15)=0.044, p=0.836, or trial x group interaction [F(1,15)= 0.476, p=0.501]. Together these results suggest that spontaneous recovery was observed but while suppression of endogenous LC activity decreased responding in previous extinction sessions, this did not translate to persistent changes in extinction learning as retention of extinction was equivalent between groups.

Discussion

The neural circuitry encoding extinction of reward-related behaviours is not fully understood. Our findings indicate that increasing activity in the LC can strengthen extinction learning. Specifically, we show that optical stimulation of the LC during extinction training significantly impacts the resilience of extinction making it resistant to spontaneous recovery, while the inhibition of LC cells with the parameters tested here did not impact the long-term expression of extinction learning.

Noradrenaline influences appetitive extinction

NA and the LC have been implicated in multiple cognitive processes and types of learning. In fear conditioning paradigms, LC neurons respond to shock and stimuli that predict shock (Sara et al., 1994; Chen and Sara 2007). Recent studies have demonstrated that a weak shock paired with optogenetic stimulation of LC axons in the basolateral amygdala (BLA) increased a conditioned fear response, mimicking the effects of a strong shock (Uematsu et al. 2017). Similarly, chemogenetic activation of LC augments fear conditioning with a weak shock as well as associated increases in BLA firing rate (Giustino et al., 2020). In contrast, optogenetic inhibition of BLA projecting LC neurons (Uematsu et al., 2017) or pharmacological blockade of BLA beta-adrenergic receptors disrupts fear conditioning (Giustino et al., 2020). This suggests that LC activity can modify the salience and/or perceived magnitude of stimuli, at least within the aversive domain. When manipulated during extinction rather than fear conditioning, optogenetic inhibition of infralimbic cortex (IFC) projecting LC neurons impaired learning, whereas inhibition of BLA-projecting neurons during extinction enhanced expression of extinction learning 24 hrs later (Uematsu et al., 2017), suggesting that LC efferents modulate fear and extinction in a pathway-specific manner.

In comparison, considerably less is known regarding the role of the LC in appetitive extinction. Appetitive paradigms provide an opportunity to study extinction learning in a situation where basal LC activity is not affected by stress or anxiety. Early studies showed 6-OHDA lesions of the dorsal noradrenergic bundle left animals resistant to extinction in an appetitive runway task (Mason & Iverson, 1975) and electrophysiological recordings showed LC activity increased following the omission of an expected appetitive reward (Sara & Segal, 1991), suggesting the LC may be involved in appetitive extinction. In the current study, using an optogenetic approach, we were able to show that LC stimulation during extinction training led to more lasting expression of extinction evidenced by reduced spontaneous recovery one and seven days after extinction learning. These findings suggest that the LC is involved in not just extinction of conditioned fear, but its role extends to appetitive extinction.

The reduction in spontaneous recovery is consistent with previous pharmacological experiments that found that atomoxetine, a NA reuptake inhibitor, also improved long-term expression of extinction despite having no obvious effect during extinction training itself (Janak & Corbit, 2011; Janak et al. 2012; Furlong et al., 2015; Leung & Corbit, 2017). It is possible that the improved retention of extinction observed here could be the result of increased in arousal or attention elicited by LC stimulation which might be expected to enhance any type of learning. It may therefore be possible that we did not enhance extinction learning by manipulating the extinction circuit, but instead activated circuitry that strengthens learning in a more general fashion (i.e., increasing attention). While we cannot rule this out entirely, it is important to note that LC stimulation specifically during stimulus presentations was important for enhancing extinction. When LC stimulation was administered for equivalent time periods during the inter-trial interval (i.e., the Offset group) there was no enhancement of extinction relative to controls suggesting the time period in which NA exerts its effect is critical for learning. The enhanced extinction observed only when LC activation coincides with reward omission makes it less likely that the observed effects were due to general increases in attention or arousal as these might have been expected for the Offset group as well. Thus, the current study builds on previous findings with the improved temporal specificity of optogenetic manipulation that could not be achieved in previous studies using lesion or pharmacological methods.

LC photoinhibition did not influence retention of extinction learning

In experiments where LC neurons were inhibited, we observed reduced responding during extinction trials when the LC was inhibited, but this did not translate to group differences in tests of short-term (i.e., the next day), or long-term (i.e., one week later) spontaneous recovery.

These results may seem at odds with previous studies that have found that administration of propranolol or inhibition of LC projections, impair extinction learning (Janak and Corbit, 2011; Janak et al., 2012; Leung & Corbit, 2017; Mueller et al., 2008; Uematsu et al., 2017). It is possible that the degree of LC basal or stimulus-evoked activation varied across these studies; for example, the LC response may be stronger in aversive paradigms and thus, the impact of LC inhibition, greater. If LC activity was already relatively low during appetitive extinction, it is possible that inhibition did not produce a big enough change in activity to result in persistent group differences in learning. It is also possible that group differences may have been detected with extinction training of shorter duration as the greatest impact of LC inhibition appeared to be in the first extinction session when response rates were highest. Despite the lower response rates observed in the eNpHR group, since rats continued to respond and experience reward omission, the extended extinction training (3 days) used here may have been sufficient to allow learning regarding the new extinction contingency in both groups.

Of note, previous pharmacological studies using appetitive paradigms have found that propranolol impairs extinction when two stimuli are extinguished together, or in compound, but not when a single stimulus is extinguished alone (Janak & Corbit, 2011). The surprising introduction of a stimulus compound generates more responding than presentations of a single stimulus, argued to relate to enhanced prediction error (Rescorla, 2006; Iordanova et al., 2021), and may better recruit LC activity, which can then be inhibited. Future studies using electrophysiological recording or fiber photometry methods should test whether this is the case.

The absence of attenuated extinction retention observed in our data could also be a result of insufficient inactivation of LC cells. While our infection rates (55 ± 8.7%) were comparable to other studies (Quinlan et al., 2019), remaining activity in the population of uninfected noradrenergic LC cells may have been sufficient to permit extinction to proceed in a manner similar to controls. It is also possible that a more precise approach targeting specific LC efferent pathways (e.g., Uematsu et al., 2017) or cell populations (Su & Cohen, 2022) may yield different results. Due to the modular organization of the LC, it is possible that global LC manipulation impacted multiple populations of cells with different forebrain targets and opposing roles with regards to extinction. Effects on competing pathways may have contributed to the lack of effect observed with inhibition and the modest effect seen with stimulation. As noted above, Uematsu et al. (2017) found that targeting LC-BLA projections affected fear learning while targeting LC-IFC projections affected extinction. As another example of modular functional organization, there were no improvements to strategy set-shifting following global LC stimulation, but improvements were observed when LC terminals in the medial prefrontal cortex were targeted (Cope et al., 2019). In addition to the proposed modular organization of the LC in terms of efferent projections, a recent electrophysiological recording study has identified two types of noradrenergic neurons within the LC and implicated just one of these (so-called "type II" neurons) in the detection of reward omission (Su & Cohen, 2022). In mice, these neurons have distinct action potential waveforms and anatomical organization which may allow them to be selectively identified and manipulated. Future experiments targeting specific efferents, such as those to the BLA and IFC, or cell populations, may refine the current results and further delineate the contribution of LC output to appetitive extinction.

LC stimulation did not produce anxiety-like or aversive behaviours

Most recent studies investigating the contributions of the LC to learning have used fear conditioning paradigms. NA is involved in stress and anxiety (Curtis et al., 2012; McCall et al., 2015; Bari et al., 2020) and direct stimulation of the entire LC may have activated circuits involved in anxiety-like or aversive behaviours (McCall et al. 2017) in addition to, or instead of, those involved in extinction. If we inadvertently activated a fear-related circuit via LC photostimulation, this could have interfered with the expression of appetitive learning because we induced a fear state rather than directly influencing appetitive learning. Our failure to find any evidence of conditioned aversion using a conditioned place preference/aversion paradigm makes conditioned aversion an unlikely explanation for our findings suggesting instead that LC stimulation, at least with the parameters tested here, specifically influenced extinction learning.

Summary

In summary, this series of experiments demonstrates that stimulation of the LC can augment reward-related extinction providing new insights into the neural circuitry that mediates appetitive extinction. Furthermore, our stimulation parameters did not elicit anxiety, indicated by the absence of any aversion in our place preference/aversion test, and our finding that stimulation outside of stimulus presentations had no effect on extinction. We observed no long-lasting changes in the retention of extinction following LC photoinhibition, suggesting that LC activity may not be necessary for extinction learning for non-aversive outcomes. Further investigations are needed to ensure that different results would not be obtained with inhibition of a higher percentage of LC cells during extinction training, conditions that generate more responding like extinction of a stimulus compound, or with selective targeting of specific LC efferent pathways.

The capacity to modify previous learning is at the heart of flexible decision making and deficits in this ability may underlie a range of neuropsychiatric disorders. Understanding the neural basis of appetitive extinction and in particular the role of the LC and NA may identify new means for enhancing extinction-based therapies.

Materials and Methods

Animals

A total of 60 Long Evans (LE) rats were included in this study. 35 of 60 (10 female, 25 male) were Long Evans-Tg(TH-Cre)3.1Deis rats (Tyrosine Hydroxylase-Cre; TH-Cre) purchased from the Rat Resource and Research Center (RRRC, Columbia, MO, USA) and the remaining 25 (5 female, 20 male) were LE rats purchased from Charles River Laboratories (CR, St. Constant, QC, Canada). As described above (**Fig.1E-G**), transgenic TH-Cre LE rats (RRRC) were confirmed to behave similarly to standard commercially available wild-type LE rats used in prior studies (CR). The robust similarities between these groups confirms, as expected based on their genetic background, that TH-Cre rats are comparable to the typical wild-type LE in their behavioural profile. Due to limited supply and problems shipping TH-Cre rats from commercial vendors during the COVID-19 pandemic, and given their similar performance in this task, a combination of TH-Cre and LE rats injected with the same control virus (AAV5-EF1a-DIO-mCherry) were collapsed for all analyses and are referred to as the Control group.

Rats were housed in pairs in individually ventilated cages in a temperature and humidity-controlled room with a 12-hr light-dark cycle (lights on at 7 AM). Training and testing took place during the light phase. Rats had unlimited access to water and environmental enrichment throughout these experiments and free access to standard laboratory chow until surgical recovery was complete. All experimental procedures were performed in accordance with guidelines from the Canadian Council on Animal Care and approved by the University of Toronto Animal Care Committee.

Stereotaxic surgery

Animals were given 1 week to acclimatize upon arrival. Once rats were at least 8 weeks old and weighed approximately 300 g, stereotaxic surgery was performed under aseptic conditions to introduce an adeno-associated virus (AAV) followed by the insertion of a dual optic implant (Doric Lenses, Quebec City, QC, Canada) into the locus coeruleus (LC). Rats were anesthetized with isoflurane (5% induction 2-5% maintenance). Meloxicam (2 mg/kg, s.c.) was administered prior to surgery for analgesia. 1.0 ml of lactated Ringer's solution with 5% dextrose was administered per hour of surgery time to improve hydration and aid in recovery. Coordinates were taken with a digital stereotaxic instrument (Kopf, Tujunga, CA, USA) and bilateral burr holes were drilled at a 20-degree angle posterior to bregma (males: A/P -12.94, M/L ±1.4, D/V -7.87; females: A/P -12.87, M/L ±1.4, D/V -7.66) to allow for the injection of the AAV. Rats were given bilateral injections of AAV5-EF1a-DIO-ChR2(H134R)-mCherry (5.2×10^12 vg/mL; ChR2 group, n=15; Offset group n= 6), AAV5-EF1a-DIO-eNpHR3.0-mCherry (4×10^12 vg/mL; eNpHR group; N=7), or AAV5-EF1a-DIO-mCherry (5.2×10^12 vg/mL; ChR2 Controls: TH-Cre LE n=7; LE n=8; Offset controls n=7; eNpHR controls n=10). All viruses were purchased from the Vector Core at the University of North Carolina. A volume of 1 µL was injected per side into the LC at 0.2 µL/min. The injection needle was left in place for 10 min before removal. Following injection, a dual optic implant (200 µm diameter optic fibres, 2.6 mm centre to centre, Doric Lenses) was implanted 0.1 mm above the injection site. Four jeweller screws on at least 2 different plates in the skull were screwed partway into the skull and these along with the implant were secured in place with dental cement (Keystone Industries, Gibbstown, NJ, USA) to make a headcap. Additional Meloxicam (2mg/kg) was administered 24 and 48 hours postoperatively. Rats were given minimum 1 week for recovery before the commencement of any behavioural training. A minimum of 3-4 weeks was allowed to pass after viral injection to allow for viral expression before optogenetic manipulations began.

Optogenetic manipulations

Optogenetic stimulation or inhibition was performed using 475 nm and 625 nm wavelength LEDs (Doric Lenses), respectively. Light output from the tip of patch cords was measured daily before the training session to ensure it was greater than 10 mW, measured using a power meter (PM100A, ThorLabs, Newton, NJ, USA). LC stimulation was administered as 5-ms pulses at 10 Hz while LC inhibition was achieved through the delivery of a constant light pulse. In both cases, light was only delivered in extinction sessions and only during stimulus presentation (20 s), or for an equivalent duration (20 s) but during the intertrial intervals (Offset group).

Behavioural training and testing

The overall goal of these experiments was to determine whether noradrenaline (NA) release originating from the LC influences extinction learning. To achieve this goal, rats were trained and then extinguished in a discriminated operant task where they earned food pellets as reward, with activity of noradrenergic LC cells manipulated at critical times during extinction.

Apparatus

Rats were trained and tested in operant chambers (Med Associates, Fairfax, VT, USA) housed within sound- and light-attenuating shells. Each chamber consists of a left and right key light situated above retractable levers and a magazine placed in the center of the same wall where a 45 mg grain-based food pellet could be delivered (BioServ, Flemington, NJ, USA; Cat# F0165). The chambers were also equipped with a white noise generator and solenoid that delivered a 5-Hz clicker stimulus that were used as auditory stimuli in addition to the visual key lights. The auditory stimuli were adjusted to 80 dB in the presence of background noise of 60 dB provided by a ventilation fan. Each chamber was illuminated by a 3-W house light. All experimental events were controlled with MED-PC V (Med Associates) software, which controlled stimulus delivery and recorded magazine entries and lever-press responses.

Magazine and lever training

After full recovery from surgery (1-2 weeks) and prior to any training, all rats were food restricted and maintained at approximately 90% of their free-feeding body weight throughout training and testing. On the first training day, rats received a 30-min magazine training session where approximately 30 pellets were delivered according to a random time 60 s schedule. The following day, rats were trained to press a lever to earn a food pellet. Each lever press delivered a pellet until 50 pellets were earned, at which point the session was terminated. Rats that failed to earn 50 pellets in the initial session were given a second identical session the following day.

Discrimination training

The next training session consisted of 24 trials (8 trials of each light, white noise and clicker stimuli). Trial order was pseudorandomly determined by the computer program and the intertrial interval (ITI) was variable but on average was 90 s. During each stimulus, a lever press would result in pellet delivery. To aid acquisition, on the first day the lever only extended during stimulus presentations and retracted when the stimulus ended. On subsequent days, the lever was present throughout the session but only responding during the stimuli was reinforced. All rats were trained on random ratio (RR) schedules that increased across days (3 days of continuous reinforcement, 3 days RR2, 3 days RR4). The rats then proceeded to the extinction sessions.

Extinction and spontaneous recovery

Each extinction session consisted of 8 presentations of one of the auditory stimuli (noise or clicker; counterbalanced) and lever-presses were recorded, but no rewards were delivered. The ITI was the same as in acquisition sessions (variable with a mean of 90 s). For the ChR2 group, during each 20 s stimulus presentation, 5 ms, 465 nm LED light pulses (Doric Lenses) were delivered at 10 Hz (Glennon et al., 2018; Quinlan et al., 2019; Vazey et al., 2018) via a patch cord. The patch cord was attached to the optic implant using ceramic connectors (Doric Lenses). The rats received 3 extinction sessions over 3 days (once per day), after which the rats were tested without stimulation for spontaneous recovery the next day and one week later. Spontaneous recovery sessions consisted of 3 trials of the same stimulus extinguished in previous sessions. There was no LED stimulation and no reward was delivered in these tests. Animals in the control group received identical treatment but lacked ChR2 and so light was expected to have no effect. Animals in the Offset group (and relevant controls) underwent identical training with the exception that stimulation in extinction sessions occurred in the middle of the variable length ITI (45s after stimulus termination, on average). Similarly, animals in the inhibition experiment were trained with near-identical parameters, with the exception that a 625 nm wavelength light was administered continuously during CS presentations in all 3 days of extinction. Testing was identical in all groups.

Conditioned Place Preference

Increased tonic LC activity has been reported to produced anxiety-like behaviour (McCall et al., 2017). For that reason, we tested rats with a conditioned place preference/aversion paradigm to determine whether LC stimulation induced anxiety-like or aversive conditioning that could influence our findings. 10 TH-Cre rats expressing ChR2 were placed in an apparatus with two chambers separated by a sliding door. One chamber, labelled Context A, had a green background with blue circles and the second chamber, labelled Context B, had a yellow background with vertical strips. All rats were given 3 days to freely explore both chambers for 10 min per day. On the fourth day, the door was closed, and 5 rats were confined to Context A where they received eight 20 s bouts of LC stimulation at 10 Hz (160 s total stimulation time) within the 10-min session. The remaining 5 rats were given the identical protocol in Context B. On alternate days, rats were confined to the opposite context where they received no stimulation. In total, rats received 3 exposures to one context with stimulation and 3 exposures to the other context without stimulation. On test day, the door was opened, and 5 rats were placed in the context paired with LC stimulation, and 5 rats placed in the context without LC stimulation. Each rat was given 10 min to freely explore both contexts and total time spent in each context was recorded. A rat was considered to be in one chamber when both hindfeet were inside the chamber.

Histology

At the end of each experiment, rats were killed with an overdose of isoflurane followed by transcardial perfusion. The rats were perfused with ice-cold 0.1 M phosphate buffered saline followed by 4% paraformaldehyde in 0.1 M phosphate buffer (PB). Brains were then extracted and stored in 4% paraformaldehyde in PB overnight and transferred to a 30% sucrose solution in 0.1 M PB for cryoprotection. Once the brains were saturated and sunk in the sucrose solution, they were frozen and sectioned with a cryostat (CM1860; Leica, Buffalo Grove, IL, USA) at 40 µm for further analysis. Immunohistochemical staining was performed to verify (1) the expression of ChR2-mCherry in TH-positive LC neurons, and (2) the location of the optic implant tip within the LC. Sections were stained for TH (mouse anti-TH, 1:1000 dilution, cat# 77-700-100, lot# 472-3JU-19, Antibodies Inc.), and mCherry (rabbit anti-mCherry, 1:1000 dilution, cat# NBP2-251-57, Novus Biologicals). Signals were enhanced with goat anti-mouse IgG Alexa Fluor 488 (1:500 dilution, RRID: AB_2534069), goat anti-rabbit IgG Alexa Fluor 568 (1:500 dilution, RRID: AB_143157). Tissue was also counterstained with DAPI in order to visualize all neurons. Sections were imaged with a confocal microscope (AxioObserver Z1; Zeiss), or under bright field (Olympus CX43). The locations of optic implant tip were plotted on plates from a rat brain atlas (Paxinos & Watson, 2007, 6th Edition).

Cell Quantification

Cells that were TH and/or mCherry positive were counted with a semi-automated program using Zen Blue (Zeiss). The accuracy of the automatic counts was verified by comparing to manual counts and the two methods found to largely overlap (90% concordance). Three representative slices were taken across the rostral/caudal axis of the LC for each animal, and this sampling strategy is based on a recent study that showed that the distribution of LC projections had no specific organization across anterior/posterior or medial/lateral axes (Schwarz et al. 2015). Each image was 0.5 x 0.5 mm and encompassed the entire structure of interest.

Statistics

Data were analyzed with mixed-model analysis of variance (ANOVA). Simple effects and post-hoc analyses were used to further assess main effects and interactions where indicated.

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Reviewer #1 (Public Review):

In this paper by Lui and colleagues, the authors examine the role of locus coeruleus (LC)-noradrenaline (NA) neurons in the extinction of appetitive instrumental conditioning. They report that optogenetic activation of global LC-NA neurons during the conditioned stimulus (CS) period of extinction enhances long-term extinction memory without affecting within-session extinction. In contrast, LC-NA activation during the intertrial interval doesn't affect extinction and long-term memory. They then show that optogenetic activation of LC-NA neurons doesn't induce conditioned place preference/avoidance. Finally, they assess the necessity of LC-NA neurons in appetitive extinction and find that optogenetic inactivation of LC-NA neurons during CS period results in enhancement of within-session extinction. The experiments are well-designed, including offset control in the optogenetic activation study. I think this study adds new insight into the LC-NA system in the context of appetitive extinction.

Strength:
・These studies identify the artificial activation of LC-NA neurons enhances long-term memory of appetitive extinction while this activation can't induce long-term conditioned place aversion. Thus, optogenetic activation of LC-NA neurons can inhibit spontaneous recovery of appetitive extinction without causing long-term aversive memory.

• Optoinhibition study demonstrates the reduction of conditioned response of within-session extinction. Therefore, LC-NA neuronal activity at the CS period of extinction could act as anti-extinction or be important for the expression of conditioned response.

Weakness:
• It is unclear how LC-NA neurons behave during the CS period of appetitive extinction from this study. This weakens the importance of the optogenetic inactivation result.
• While authors manipulate global LC-NA neurons, many people find functionally heterogeneous populations in the LC. It remains unsolved if there is specific LC-NA subpopulation responsible for appetitive extinction.

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Reviewer #2 (Public Review):

Understanding how the LC/noradrenaline system controls basic cognitive processes is important and timely. This study aims to understand the role Locus Coeurelus /noradrenaline system in extinction of conditioned responding. The authors used a discriminative appetitive procedure to show that photoexcitation of noradrenergic neurons of the Locus Coeruleus has no effect on the performance during extinction but impacts expression of extinguished responding through a decreased spontaneous recovery. This study is appropriately designed and the results are well analysed. Therefore, it provides an important and timely addition to the field

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Reviewer #3 (Public Review):

The introduction/background is excellent. It reviews evidence showing that extinction of conditioned responding is regulated by noradrenaline and suggesting that the locus coeruleus (LC) may be a critical locus of this regulation. This naturally leads to the aim of the study: to determine whether the locus coeruleus is involved in extinction of an appetitive conditioned response. Overall, the study is well designed, nicely conducted and the results advance our understanding of the role of the LC in extinction of conditioned behaviour. Future studies may provide more fine-grained analyses of behavioral data to clarify the impact of the LC manipulations (stimulation and inhibition) on performance in the task.

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Author Response

The following is the authors' response to the current reviews.

We thank the editors and reviewers for their careful consideration of our revised manuscript. Reviewers 2 and 3 indicated that their previous comments had been satisfactorily addressed by our revisions. Reviewer 1 raised several points and our point by point responses can be found below.

> ***Response to Reviewer Comments:***

We thank the editors and reviewers for their careful consideration of our revised manuscript. Reviewers 2 and 3 indicated that their previous comments had been satisfactorily addressed by our revisions. Reviewer 1 raised several points and our point by point responses can be found below.

> ***Reviewer #1 (Recommendations For The Authors):***
>
> 1. *Please clarify the terminology of spontaneous recovery in your study.*
>
>   *According to Rescorla RA 2004 (http://www.learnmem.org/cgi/doi/10.1101/lm.77504.), he defines spontaneous recovery as "with the passage of time following nonreinforcement, there is some "spontaneous recovery" of the initially learned behavior. ". So in this study, I thought Test2 is spontaneous recovery while the Test1 is extinction test as most studies do. But authors seem to define spontaneous recovery from the last trial of Extinction3 to the first trial of Test1, which is confusing to me.*

We agree with the reviewer (and Rescorla, 2004) that spontaneous recovery is defined as the return of the initially learned behaviour after the passage of time. In our study, Test 1 is conducted 24-hours after the final extinction session (Extinction 3) and in our view, the return of responding following that 24-hour delay can be considered spontaneous recovery. Rescorla (2004 and elsewhere) also points out that the magnitude of spontaneous recovery may be greater with larger delays between extinction and testing. This in part motivated our second test 7 days following the last extinction session with optogenetic manipulation. We did not find evidence of greater spontaneous recovery in the test 7 days later, however, the additional extinction trials in Test 1 may have reduced the opportunity to detect such an effect.

> 1. *Why are E6-8 plots of Offset group in Figure 3E and F different?*

We apologise for this error and have corrected it. This was an artifact of an older version of the figure before final exclusions. The E6-8 data is now the same for panels 2E and 2F.

> 1. *Related to 2, Please clarify what type of data they are in Figure3E,F Figure5H, and I . If it's average, please add error bars. Also, it's hard to see the statistical significance at the current figure style.*

The data in these panels are the mean lever presses per trial as labeled on the y-axis of the figures. In our view, in this instance, error bars (or lines and other markers of significance) detract from the visual clarity of the figure. The statistical approach and outcomes are included in the figure legend and when presented alongside the figure in the final version of the paper should directly clarify these points.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *The authors have addressed my previous comments to my satisfaction.*

The authors have adequately addressed each of the points raised in my original review. The paper will make a nice contribution to the field.

> ***Reviewer #3 (Recommendations For The Authors):***

The authors have adequately addressed each of the points raised in my original review. The paper will make a nice contribution to the field.

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Recommendations For The Authors):***
>
> - *It would be interesting if the authors would do calcium imaging or electrophysiology from LCNA neurons during appetitive extinction.*

Indeed these are interesting ideas. We have plans to pursue them but ongoing work is not yet ready for publication.

> - *LC-NA neuronal responses during the omission period seem to be important for appetitive extinction as described in the manuscript (Park et al., 2013; Sara et al., 1994; Su & Cohen 2022). It would be nice to activate/inactivate LC-NA neurons during the omission period.*

Optogenetic manipulation was given for the duration of the stimulus (20 seconds; when reward should be expected contingent upon performance of the instrumental response). We believe the reviewer is suggesting briefer manipulation only at the precise time the pellet would have been expected but omitted. If so, the implementation of that is complex because animals were trained on random ratio schedules and so when exactly the pellet(s) was earned was variable and so when precisely the animal experiences "omission" is difficult to know with better temporal specificity than used in the current experiments. But we agree with the reviewer that now we see that there is an effect of LC manipulation, in future studies we could alter the behavioral task so that the timing of reward is consistent (e.g., train the animals with fixed ratio schedules or continuous reinforcement, or use a Pavlovian paradigm) where a reasonable assertion about when the outcome should occur, and thus when its absence would be detected, can be made and then manipulation given at that time to address this point.

> - *Does LC-NA optoinhibition affect the expression of the conditioned response (the lever presses at early trials of Extinction 1)? It's hard to see this from the average of all trials.*

The eNpHR group responded numerically less overall during extinction. This effect appears greatest in the first extinction session, but fails to reach statistical significance [$F(1,15)= 3.512$, $p=0.081$]. Likewise, analysis of the trial by trial data for the first extinction session failed to reveal any group differences [$F(1,15)= 3.512$, $p=0.081$] or interaction [trial x group; $F(1,15)=0.550$, $p=0.470$].

Comparison of responding in the first trial also failed to reveal group differences [$F(1.15)=1.209$, $p=0.289$]. Thus while there is a trend in the data, this is not borne out by the statistical analysis, even in early trials of the session.

> - *While the authors manipulate global LC-NA neurons, many people find the heterogeneous populations in the LC. It would be great if the authors could identify the subpopulation responsible for appetitive extinction.*

We agree that it would be exciting to test whether and identify which subpopulation(s) of cells or pathway(s) are responsible for appetitive extinction. While related work has found that discrete populations of LC neurons mediate different behaviours and states, and may even have opposing effects, our initial goal was to determine whether the LC was involved in appetitive extinction learning. These are certainly ideas we hope to pursue in future work.

> *Minor:*
>
> - *Why do the authors choose 10Hz stimulation?*

The stimulation parameters were based on previously published work. We have added these citations to the manuscript.

Quinlan MAL, Strong VM, Skinner DM, Martin GM, Harley CW, Walling SG. Locus Coeruleus Optogenetic Light Activation Induces Long-Term Potentiation of Perforant Path Population Spike Amplitude in Rat Dentate Gyrus. Front Syst Neurosci. 2019 Jan 9;12:67. doi: 10.3389/fnsys.2018.00067. PMID: 30687027; PMCID: PMC6333706.

Glennon E, Carcea I, Martins ARO, Multani J, Shehu I, Svirsky MA, Froemke RC. Locus coeruleus activation accelerates perceptual learning. Brain Res. 2019 Apr 15;1709:39-49. doi: 10.1016/j.brainres.2018.05.048. Epub 2018 May 31. PMID: 29859972; PMCID: PMC6274624.

Vazey EM, Moorman DE, Aston-Jones G. Phasic locus coeruleus activity regulates cortical encoding of salience information. Proc Natl Acad Sci U S A. 2018 Oct 2;115(40):E9439-E9448. doi: 10.1073/pnas.1803716115. Epub 2018 Sep 19. PMID: 30232259; PMCID: PMC6176602.

> - *The authors should describe the behavior task before explaining Fig1e-g results.*

We agree that introducing the task earlier would improve clarity and have added a brief summary of the task at the beginning of the results section (before reference to Figure 1) and point the reader to the schematics that summarize training for each experiment (Figures 2A and 4D).

NOTE R2 includes specific comments in their Public review. We have considered those as their recommendations and address them here.

> 1. *In such discrimination training, Pavlovian (CS-Food) and instrumental (LeverPress-Food) contingencies are intermixed. It would therefore be very interesting if the authors provided evidence of other behavioural responses (e.g. magazine visits) during extinction training and tests.*

In a discriminated operant procedure, the DS (e.g. clicker) indicates when the instrumental response will be reinforced (e.g., lever-pressing is reinforced only when the stimulus is present, and not when the stimulus is absent). This is distinct from something like a Pavlovianinstrumental transfer procedure and so we wish to just clarify that there is no Pavlovian phase where the stimuli are directly paired with food. After a successful lever-press the rat must enter the magazine to collect the food, but food is only delivered contingency upon lever-pressing and so magazine entries here are not a clear indicator of Pavlovian learning as they may be in other paradigms.

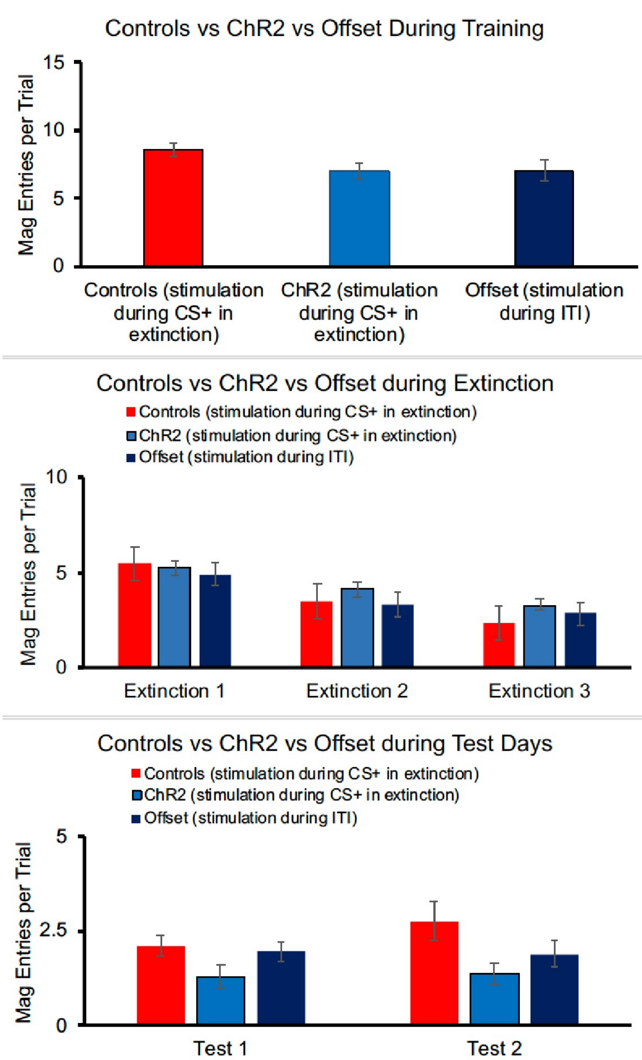
Nonetheless, we have compiled magazine entry data which although not fully independent of the lever-press response in this paradigm, still tells us something about the animals' expectation regarding reward delivery.

For the ChR2 experiment, largely paralleling the results seen in the lever-press data, there were no group differences in magazine responses at the end of training [F(2,40)=2.442, p=0.100].

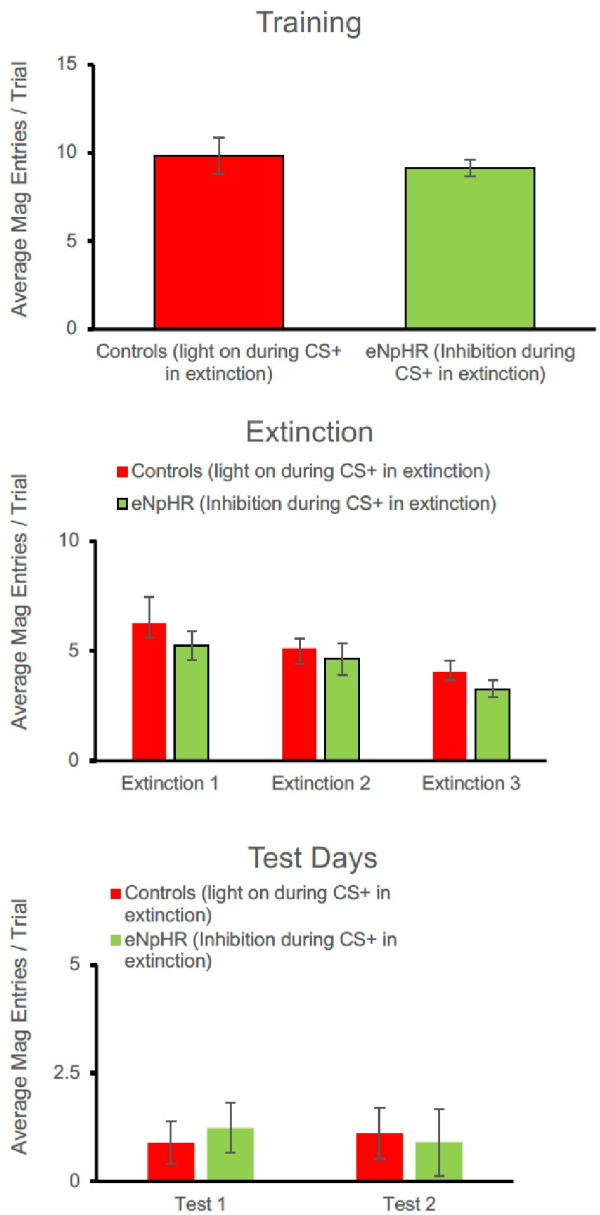
Responding decreased across days of extinction (when optogenetic stimulation was given) [F(2, 80)=38.070, p<0.001], but there was no effect of group [F(2,40)=0.801, p=0.456] and no interaction between day and group [F(4,40)=1.461, p=0.222]. Although a similar pattern is seen in the test data, group differences were not statistically different in the first [F(2,40)=2.352, p=0.108] or second [F(2,40)=1.900, p=0.166] tests, perhaps because magazine responses were quite low. Thus, overall, magazine data do not present a different picture than lever-pressing, but because of the lack of statistical effects during testing, we have chosen not to include these data in the manuscript.

For the eNpHR experiment, again a similar pattern to lever-pressing was seen. There were no group differences at the end of acquisition [F(1,15)=0.290, p=0.598]. Responding decreased across days of extinction [F(2, 30)=4.775, p=0.016] but there was no main effect of group [F(1,15)=1.188, p=0.293], and no interaction between extinction and group [F(2,30)=0.070, p=0.932]. There were no group differences in the number of magazine entries in Test 1 [F(1,15)=1.378, p=0.259] or Test 2 [F(1,15)=0.319, p=0.580].

Author response image 1.



Author response image 2.



> 1. *In Figure 1, the authors show the behavioural data of the different groups of control animals which were later collapsed in a single control group. It would be very nice if the authors could provide the data for each step of the discrimination training.*

We are a little confused by this comment. Figure 1, panels E, F, and G show the different control groups at the end of training, for each day of extinction (when manipulations occurred) and for each test, respectively. It's not clear if there is an additional step the reviewer is interested in? We note neural manipulation only occurred during extinction sessions.

We chose to compare the control groups initially, and finding no differences, to collapse them for subsequent analyses as this simplifies the statistical analysis substantially; when group differences are found, each of the subgroups has to be investigated (including the different controls means there are 5 groups instead of 3). It doesn't change the story because we tested that there were not differences between controls before collapsing them, but collapsing the controls makes the presentation of the statistical data much shorter and easier to follow.

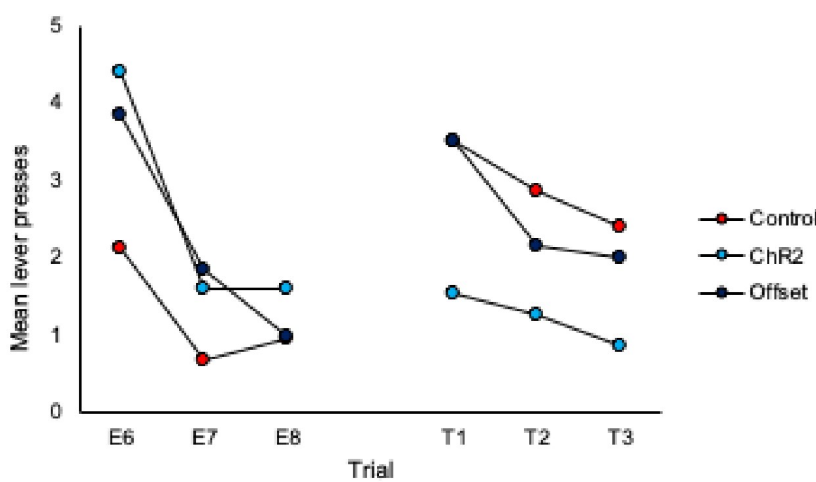
> 1. *Inspection of Figures 2C & 2D shows that responding in control animals is about the same at test 2 as at the end of extinction training. Therefore, could the authors provide evidence for spontaneous recovery in control animals? This is of importance given that the main conclusion of the authors is that LC stimulation during extinction training led to an increased expression of extinction memory as expressed by reduced spontaneous recovery.*

To address this we have added analyses of trial data, specifically comparison of the final 3 trials of extinction to the subsequent three trials of each test. These analyses are included on page 5 of the manuscript and additional data figures can be found as panels 2E and 2F and pasted below.

What we observe in the trial data for controls is an increase in responding from the end of extinction to the beginning of each test, thus demonstrating spontaneous recovery. Importantly, responding in the ChR2 group does not increase from the end of extinction to the beginning of the test, illustrating that LC stimulation during extinction prevents spontaneous recovery.

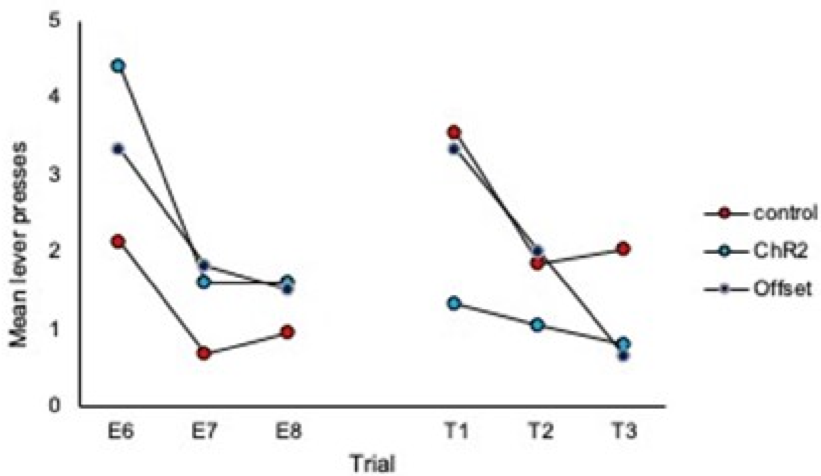
Comparison of the final three trials of Extinction to the three trials of Test 1:

Author response image 3.



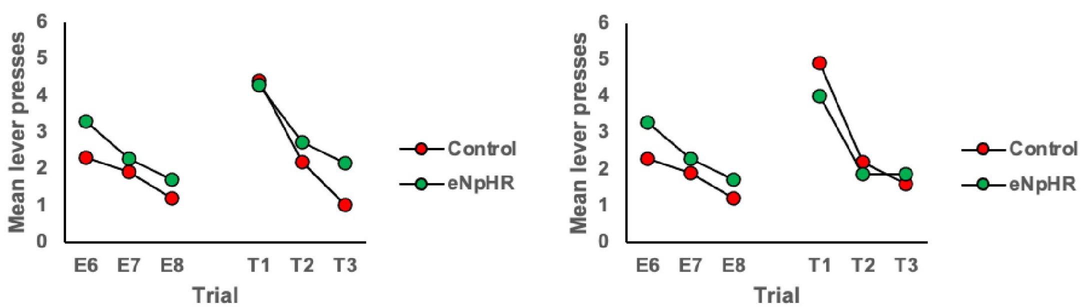
Comparison of the final three trials of Extinction to the three trials of Test 2:

Author response image 4.



Halorhodopsin Experiment Tests 1 and 2, respectively.

Author response image 5.



> 1. *Current evidence suggests that there are differences in LC/NA system functioning between males and females. Could the authors provide details about the allocation of male and female animals in each group?*

More females had surgical complications (excess bleeding) than males resulting in the following allocations; control group; 14 males and 8 females; ChR2 group 8 males and 7 females; offset 6 males.

In our dataset, we did not detect sex differences in training [no main effect of sex: F(1,38)=1.097, p=0.302, sex x group interaction: F(1,38)= 1.825, p=0.185], extinction [no effect of sex; F(1,38)=0.370, p=0.547; no sex x extinction interaction: F(2,76)=0.701, p=0.499 ; no sex x extinction x group interaction: F(2,76)=2.223, p=0.115] or testing [Test 1 no effect of sex: F(1,38)=1.734, =0.196; no sex x group interaction: F(1,38)=0.009, p=0.924; Test 2 no effect of sex: F(1,38)=0.661, p=0.421; no sex x group interaction: F(1,38)=0.566, p=0.456].

1. *The histology section in both experiments looks a bit unsatisfying. Could the authors provide more details about the number of counted cells and also their distribution along the anteroposterior extent of the LC. Could the authors also take into account the sex in such an analysis?*

The antero-posterior coordinates used for cell counts and calculation of % infection rates were between -9.68 and -10.04 (Paxinos and Watson, 2007, 6th Edition) as infection rates were most consistent in this region and it was well-positioned relative to the optic probe although TH and mCherry positive cells were observed both rostral and caudal to this area. For each animal, an average of ~116+/- 25 TH-positive LC neurons as determined by DAPI and GFP positive cells were identified. Viral expression was identified by colocalized mCherry staining. Animals that did not have viral expression in the LC were not included in the experimental groups. We have added these details to the histology results on page 4.

Males and females showed very similar infection rates (Males, 74%; Females, 72%). While sex differences, such as total number of LC cells or total LC volume have been reported (Guillamon, A. et al. 2005), Garcia-Falgueras et al. (2005) reported no differences in LC volume or number of LC neurons between male and female Long-Evans rats. So while differences may exist in the LC of Long-Evans rats, the cell counts here were comparable between groups (males, 103 +/- 27; females, 129 +/- 17; t-test, p>0.05).

References:

1. Garcia-Falgueras, A., Pinos, H., Collado, P., Pasaro, E., Fernandez, R., Segovia, S., & Guillamon, A. (2005). The expression of brain sexual dimorphism in artificial selection of rat strains. Brain Research, 1052(2), 130–138. https://doi.org/10.1016/j.brainres.2005.05.066

2. Guillamon, A., De Bias, M. R., & Segovia, S. (1988). Effects of sex steroids on the of the locus coeruleus in the rat. Developmental Brain Research, 40, 306–310.

Reviewer #3 (Recommendations For The Authors):

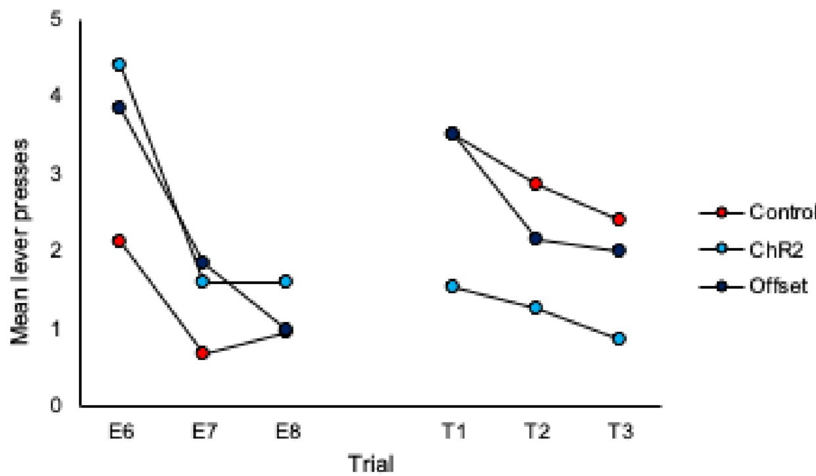
MAJOR

1. *It is worth noting that responding in Group ChR2 decreased from Extinction 3 to Test 1, while responding in the other two groups appears to have remained the same. This suggests that there was no spontaneous recovery of responding in the controls; and, as such, something more must be said about the basis of the between-group differences in responding at test. This is particularly important as each extinction session involved eight presentations of the to-betested stimulus, whereas the test itself consisted of just three stimulus presentations. Hence, comparing the mean levels of performance to the stimulus across its extinction and testing overestimates the true magnitude of spontaneous recovery, which is simply not clear in the results of this study. That is, it is not clear that there is any spontaneous recovery at all and, therefore, that the basis of the difference between Group ChR2 and controls at test is in terms of spontaneous recovery.*

The reviewer is correct that there were a different number of trials in extinction vs. test sessions making direct comparison difficult and displaying the data as averages of the test session does not demonstrate spontaneous recovery per se. To address this we have added analyses of trial data and comparison of the final 3 trials of extinction to the subsequent three trials of each test. These analyses are included on page 5 and 6 of the manuscript and additional data figures can be found as panels 2E and 2F and 4 H and I, and pasted below. What we observe in the trial data for controls is an increase in responding from the end of extinction to the beginning of each test, thus demonstrating spontaneous recovery. Importantly, responding in the ChR2 group does not increase from the end of extinction to the beginning of the test, illustrating that LC stimulation during extinction prevents spontaneous recovery.

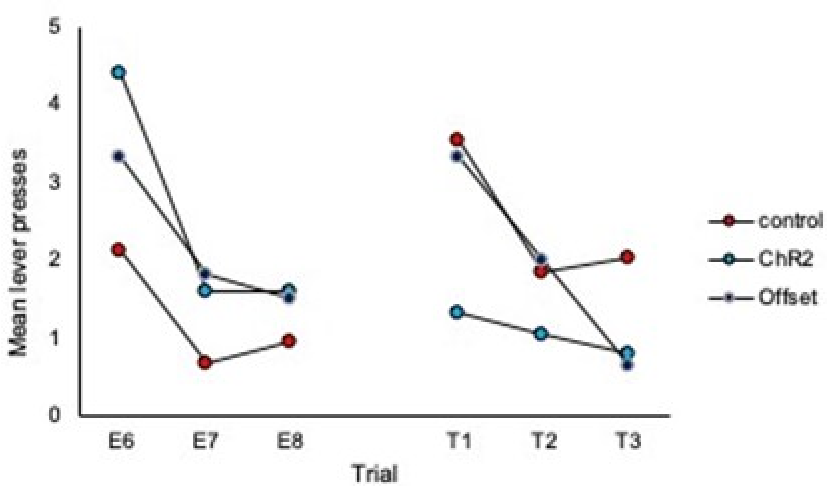
Comparison of the final three trials of Extinction to the three trials of Test 1:

Author response image 6.



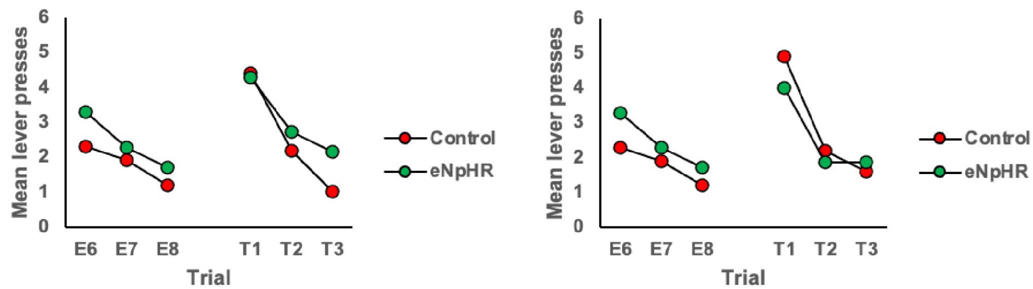
Comparison of the final three trials of Extinction to the three trials of Test 2:

Author response image 7.



Halorhodopsin Experiment Tests 1 and 2, respectively.

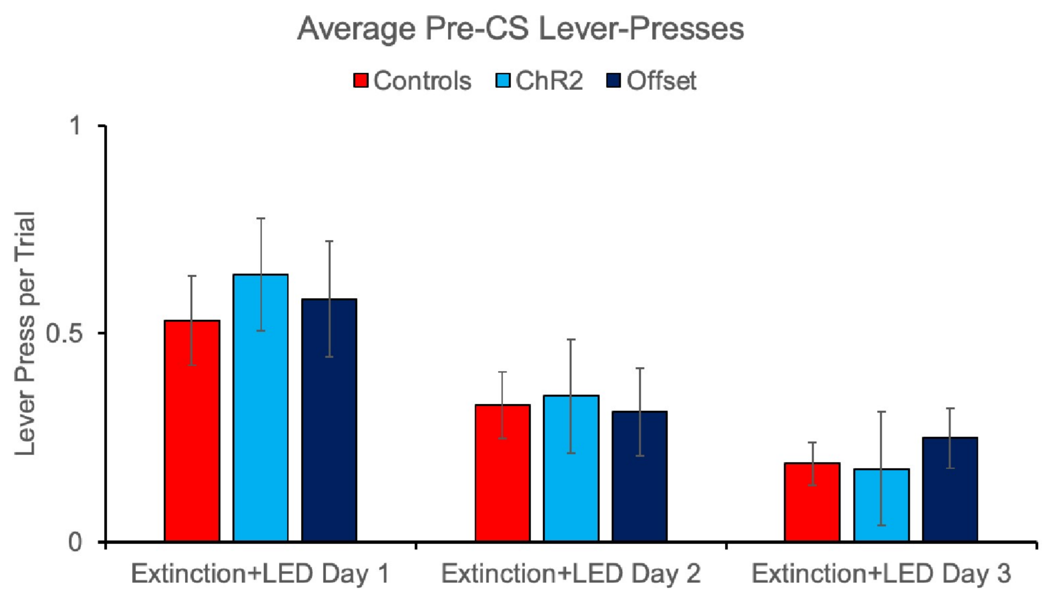
Author response image 8.



> *2a) Did the manipulations have any effect on the rates of lever-pressing outside of the stimulus?*

We did not detect any effect of the optogenetic manipulations on rates of lever pressing outside of the stimulus. This is demonstrated in the pre-CS intervals collected on stimulation days (i.e., extinction sessions) where we see similar response rates between controls and the ChR2 and Offset groups as shown below. There was no effect of group [F(2,40)=0.156, 0.856] or group x extinction day interaction [F(2,40)=0.146, p=0.865].

Author response image 9.

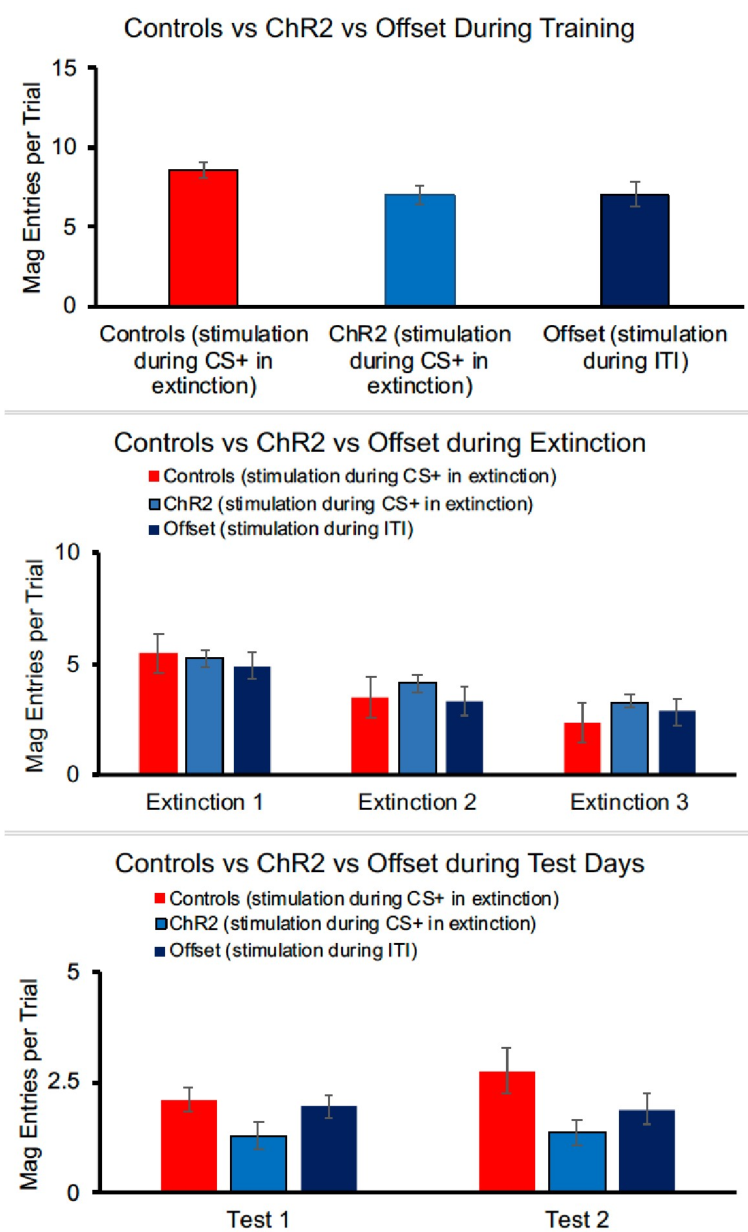


> *2b) Did the manipulations have any effect on rates of magazine entry either during or after the stimulus?*

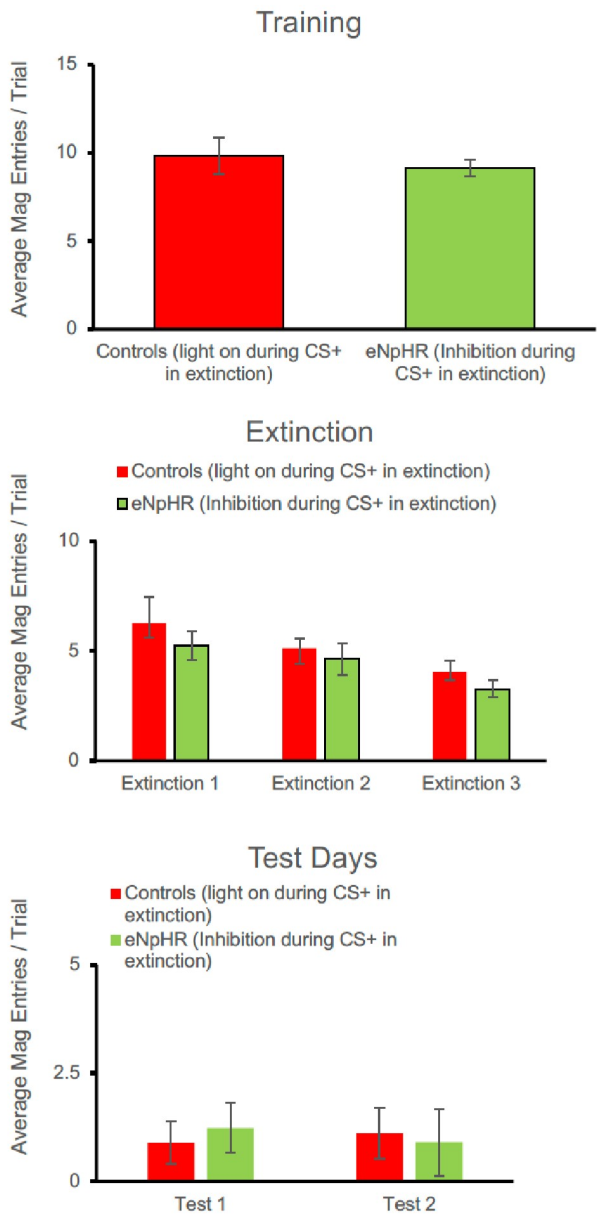
For the ChR2 experiment, there were no group differences in magazine responses at the end of training [F(2,40)=2.442, p=0.100]. Responding decreased across days of extinction (when optogenetic stimulation was given) [F(2, 80)=38.070, p<0.001], but there was no effect of group [F(2,40)=0.801, p=0.456] and no interaction between day and group [F(4,40)=1.461, p=0.222]. Although a similar pattern is seen in the test data, group differences were not statistically different in the first [F(2,40)=2.352, p=0.108] or second [F(2,40)=1.900, p=0.166] tests, perhaps because magazine responses were quite low. Thus, overall, magazine data do not present a different picture than lever-pressing, but because of the lack of statistical effects during testing, we have chosen not to include these data in the manuscript.

For the eNpHR experiment, again a similar pattern to lever-pressing was seen. There were no group differences at the end of acquisition [F(1,15)=0.290, p=0.598]. Responding decreased across days of extinction [F(2, 30)=4.775, p=0.016] but there was no main effect of group [F(1,15)=1.188, p=0.293], and no interaction between extinction and group [F(2,30)=0.070, p=0.932]. There were no group differences in the number of magazine entries in Test 1 [F(1,15)=1.378, p=0.259] or Test 2 [F(1,15)=0.319, p=0.580].

Author response image 10.



Author response image 11.



> *2c) Did the manipulations affect the coupling of lever-press and magazine entry responses? I imagine that, after training, the lever-press and magazine entry responses are coupled: rats only visit the magazine after having made a lever-press response (or some number of leverpress responses). Stimulating the LC clearly had no acute effect on the performance of the lever-press response. If it also had no effect on the total number of magazine entries performed during the stimulus, it would be interesting to know whether the coupling of lever-presses and magazine entries had been disturbed in any way. One could assess this by looking at the jointdistribution of lever-presses (or runs of lever-presses) and magazine visits in each extinction session, or across the three sessions of extinction. As a proxy for this, one could look at the average latency to enter the magazine following a lever-press response (or run of leverpresses). Any differences here between the Controls and Group ChR2 would be informative with respect to the effects of*

> *the LC manipulations: that is, the results shown in Figure indicate that stimulating the LC has no acute effects on lever-pressing but protects against something like spontaneous recovery; whereas the results shown in Figure 4 indicate that inhibiting the LC facilitates the loss of responding across extinction without protecting against spontaneous recovery. The additional data/analyses suggested here would indicate whether LC stimulation had any acute effects on responding that might explain the protection from spontaneous recovery; and whether LC inhibition specifically reduced lever-pressing across extinction or whether it had equivalent effects on rates of magazine entry.*

Lever-press and magazine response data were collected trial by trial but not with the temporal resolution required for the analyses suggested by the reviewer. We do not have timestamps for magazine entries nor latency data. We can collect this type of data in future studies. At the session or trial level, magazine entries generally correspond to lever-pressing; being trained on ratio schedules, and from informal observation, rats will do several lever-presses and then check the magazine. Rates of each decrease across extinction (magazine data included in response to comment 2b. above). Optogenetic manipulation appeared to have no immediate effect on either response during extinction.

> *ROCEDURAL*
>
> 1. *Why were there three discriminative stimuli in acquisition: a light, white noise, and clicker?*

This was done to be consistent with and apply parameters similar to previous, related studies (Rescorla, 2006; Janak & Corbit, 2011) and to allow comparison to potential future studies that may involve stimulus compounds etc. (requiring training of multiple stimuli).

> 1. *Why were some rats extinguished to the noise while others were extinguished to the clicker? Were the effects of LC stimulation/inhibition dependent on the identity of the extinguished stimulus?*

Because the animals were trained with multiple stimuli, it allowed us some ability to choose amongst those stimuli to best balance response rates across groups before the key manipulations. The effects of LC manipulation did not differ between animals based on the identity of the extinguished stimulus.

> 1. *Did the acute effects of LC inhibition on extinction vary as a function of the stimulus identity?*

No

> 1. *Was the ITI in extinction the same as that in acquisition?*

Yes, the ITI was the same for acquisition and extinction sessions (variable, averaging to 90 seconds). We have added a sentence to the methods (p. 11) to reflect this.

> 1. *For Group Offset, when was the photo-stimulation applied in relation to the extinguished stimulus: was it immediately upon offset of the stimulus or at a later point in the ITI?*

The group label “Offset” was used to be consistent with Umaetsu et al. (2017) that delivered stimulation 50-70s after a trial. SImilarly, we mean it as discontinuous with the stimulus, not at the termination of the stimulus. We have revised the description of this group on page 11 to clarify the timing of the photostimulation as follows:

“Animals in the Offset group (and relevant controls) underwent identical training with the exception that stimulation in extinction sessions occurred in the middle of the variable length ITI (45s after stimulus termination, on average).”

> *MINOR*
>
> 1. *"Such recovery phenomena undermine the success of extinction-based therapies..."*

***Perhaps a different phrasing is needed here: "These phenomena show that extinction-based therapies are not always effective in suppressing an already-established response..."

We have revised this sentence in line with the reviewer’s suggestion:

“These phenomena mean that extinction-based therapies are not always successful in suppressing previously-established behaviours” (first paragraph of the introduction).

> 1. *Typo in para 1 of results: "F(2,19)=0.0.352"*

Thank you for finding this typo. It has been corrected. (p.4)

> 1. *"As another example of modular functional organization, no improvements to strategy setshifting following global LC stimulation, but improvements were observed when LC terminals in the medial prefrontal cortex were targeted (Cope et al., 2019)." ***This sentence is missing a "there were" before "no improvements".*

Thank you for finding this error. It has been corrected. (p.8)